BAES, FRANZ EMMANUEL F.
BSIT 2-1

COMP-009 -OOP
3/2/26

Part A- Variable Identification Table

Variable Name	DATA TYPE	VARIABLE TYPE	STORED IN
name	STRING	INSTANCE	HEAP
grade	INT	INSTANCE	HEAP
S1	Student	LOCAL	STACK (Reference) HEAP (OBJECT)
S2	Student	LOCAL	STACK (reference) HEAP (OBJECTS
S3	Student	LOCAL	STACK (reference) HEAP (OBJECT)

PART B- Explanation. Please see example below

STATEMENT	WHAT HAPPENS?	MEMORY IMPACT
Student s1 = new Student("John", 85);	A Student object is created with name = "John" and grade = 85 which is then stored to s1 as reference.	New object on Heap, s1 on stack points to it.
Student s2 = new Student("Maria", 90);	second Student object is created with name = "Maria" and grade = 90. s2 stores the reference.	New object created on Heap. s2 stack points to it.
Student s3 = s2;	There will be no object. s3 copies s2's reference	No Heap memory used s3 stack points to the same object as s2
s1.grade = 95;	grade of object (John) is updated to 95	John's object on heap is modified
s3.~~name~~ name = "Ana";	since s3 and s2 points to the same object this changes the name from "Maria" to "Ana" for Both	Maria's object in the heap is modified s2 ~~and~~ s3 become "Ana"

Part C : OUTPUT

John 95
Ana 90
Ana 90

Scanned with CamScanner